

Problem 1: Show that

$$\prod_{\substack{p \leq n \\ p \text{ is prime}}} p < 4^n$$

for every natural number n greater than or equal to two.

Solution:

Let n be a natural number greater than or equal to two. The product of all primes less than or equal to n is necessarily less than $n^{\pi(n)}$, where π is the prime-counting function. $\pi(n)$ is usually defined as $\pi(n) = \frac{n}{\ln(n)}$, but we can use $\frac{1.26n}{\ln(n)}$ as an upper bound. So our bound can be written as $n^{\frac{1.26n}{\ln(n)}}$.

We can use the change-of-base formula for an exponential function.

$$n^{\frac{1.26n}{\ln(n)}} = e^{\ln(n) \frac{1.26n}{\ln(n)}} = e^{1.26n}$$

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Thus our bound can be written more simply as $e^{1.26n}$.

$$\prod_{\substack{p \leq n \\ p \text{ is prime}}} p < e^{1.26n}$$

Since $e^{1.26n} < 4^n$, it follows that

$$\prod_{\substack{p \leq n \\ p \text{ is prime}}} p < 4^n$$

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